

Gursimran Singh

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Senior Software Engineer with 7+ years of experience designing, developing, and delivering scalable web applications across frontend and full-stack technologies. Hands-on experience building AI-powered applications using OpenAI APIs, Retrieval-Augmented Generation (RAG), vector search, Hugging Face, and function calling. Proven success in delivering high-performance, user-centric solutions and driving business value through modern software engineering practices.

PROFESSIONAL EXPERIENCE

Trantor Inc

Senior Software Engineer

Chandigarh, India

Sept 2021–Present

- **Led end-to-end product development** for multiple web applications transforming business requirements into scalable, production-ready solutions while collaborating closely with stakeholders throughout the development lifecycle.
- **Engineered and optimized high-traffic web applications** across logistics, recruitment, and digital marketing domains, including a campaign platform that successfully served **1.2M+ users within a one-hour window while sustaining 120 TPS (transactions per second)** with high availability and performance.
- **Collaborated with product owners, designers, and clients** to perform feature grooming, feasibility analysis, technical planning, code reviews, and seamless implementation based on Figma designs and business requirements.
- **Maintained engineering excellence** by implementing CI/CD pipelines, writing unit tests, integrating third-party services and iframes, developing scheduled cron jobs, ensuring SonarQube code quality standards, and following software development best practices.

Aristeia services

Senior Software Engineer

Singapore

Feb 2021–Sept 2021

- **Developed and maintained content-rich blogging platforms and social sharing tools using React.js**, delivering responsive, pixel-perfect user interfaces from Figma/mockups while implementing new features using Agile development methodologies.
- **Collaborated with cross-functional teams** to build reusable UI components, integrate APIs, manage Docker-based development environments, and contribute to CI/CD pipelines, ensuring reliable deployments and high-quality software delivery.

Agile Media Lab

Software Engineer

Chandigarh, India

Nov 2018–Jan 2021

- **Modernized legacy applications by migrating Backbone.js modules to React.js**, improving code maintainability, reusability, and overall application performance while ensuring a seamless user experience.
- **Delivered day-to-day product enhancements** by developing and integrating REST APIs, implementing new features, resolving production issues and bugs, and collaborating with backend teams to ensure reliable and scalable application functionality.

EDUCATION

Panjab University

Master of Computer Applications (MCA)

72.6%

Chandigarh

May 2019

Kurukshetra University

Bachelor of Science (B.Sc.)

75%

Kurukshetra, India

May 2016

H.B.S.E

XII (Non-Medical)

82.6%

Bhiwani, Haryana, India

May 2013

PROJECTS

Tevnt

<https://tevnt.com>, [chaseSapphire-visapromotion](#), [visafwcscoretickets](#)

Tech Stack: AstroJS, ReactJS, i18n, Tailwind, TypeScript

- **Developed and delivered multiple high-traffic FIFA World Cup campaign websites for Visa, Chase, Wells Fargo, and Capital One using React.js, Astro.js, Tailwind CSS, Axios, and React Testing Library**, consistently meeting aggressive delivery timelines.
- Built high-performance campaign applications handling **1.2M+ users within a one-hour window** while sustaining **~120 TPS**.
- Integrated **Visa iframes** and delivered **~20,000 secure ticket redemption links** with retry logic and campaign state management.
- Implemented **GTM, cookie consent, and compliance requirements**, maintaining **zero Checkmarx security findings**.
- Converted **Figma designs into pixel-perfect, responsive UIs** with reusable components and comprehensive testing.

AI Todo Management Assistant

Tech Stack: Node JS, OpenAI, TypeScript

- Built an AI-powered Todo Assistant using **OpenAI Function Calling** to create, update, prioritize, complete, and retrieve tasks through natural language.
- Designed structured tool schemas, conversation management, and contextual prompting for reliable task execution.
- Implemented date normalization, structured outputs, and intelligent intent recognition to automate task management workflows.
- Implemented OpenAI tool calling, structured outputs, and conversation state management to execute reliable multi-step task workflows.

Browser Based RAG Assistant

Tech Stack: React JS, Transformers.JS, Web workers, Small LLM Models

- Developed a browser-based document Q&A assistant using **Transformers.js, Web Workers, and Retrieval-Augmented Generation (RAG)**.
- Implemented document chunking, embedding generation, cosine similarity search, and context-aware answer generation entirely on the client side.
- Optimized model loading, caching, and background inference to deliver a responsive user experience without blocking the UI thread.
- Built browser-side inference using Web Workers to prevent UI blocking during embedding generation and text generation.

AI Document Search System

Tech Stack: NestJS, PostgreSQL, pinecone, LangChain

- Designed a scalable document search system using **vector embeddings**, **pinecone**, and semantic similarity search.
- Implemented document ingestion, embedding generation, project-based access control, and retrieval pipelines for enterprise knowledge search.

Source-Smart

ss.trantorinc.com

Tech Stack: Antd, React JS, TypeScript, Express Js,

- **Built responsive, pixel-perfect user interfaces** using **React.js**, **Ant Design**, **React Router**, and **Axios**, translating Figma designs into production-ready features.
- **Implemented core application features** including document upload (PDF/DOCX), resume downloads, role-based access control (RBAC), candidate matching logic, and FSR interview question generation.
- **Optimized application performance** through lazy loading, code splitting, and bundle size optimization, ensuring a fast and responsive user experience.
- **Worked in an Agile environment**, participating in requirement discussions, feature planning, API integrations, code reviews, bug fixing, and production deployments.
- Developed an automated recruitment workflow which does resume shortlisting based on job descriptions, significantly improving the efficiency of the hiring process

Customer Discovery Tool

cdt.vmware.com

Tech Stack: React JS, React Router, Jest, Enzyme

- **Developed and maintained an enterprise admin application** using **React.js**, delivering modern, responsive, and pixel-perfect user interfaces based on Figma designs.
- **Optimized application performance** by implementing caching strategies, efficient rendering techniques, and handling large datasets for improved responsiveness.
- **Integrated REST APIs** and developed reusable, scalable components while collaborating in **Agile/Scrum** teams to deliver business-critical features.
- **Implemented unit tests** using **Jest** and **Enzyme**, ensuring application reliability and maintainable code quality
- **Contributed to CI/CD pipelines** and worked with **Docker/Kubernetes-based deployment environments** to support seamless application releases

Webedia-Espana

<https://xataka.com>, <https://tendencias.com>, directoalpaladar.com

Tech Stack: Php Symfony, Node Js, Express JS, React, Next JS, Docker

- **Modernized a large-scale publishing platform** by migrating legacy **Backbone.js** modules to **React.js**, improving maintainability, performance, and user experience across multiple websites sharing a single codebase.
- **Developed and integrated REST APIs** while contributing to **Node.js** and **PHP Symfony** microservices, implementing cron jobs and backend functionalities to support business-critical workflows.

- **Improved application performance and SEO by implementing Varnish caching, optimizing page rendering, and supporting monetization through Google Ads and Header Bidding integrations.**
- **Built pixel-perfect, responsive user interfaces** from Figma/mockups, collaborating with designers and product teams to deliver production-ready features.
- **Implemented and managed cloud infrastructure** using **AWS EC2, ECS, Docker, SNS, and SQS**, contributing to scalable deployments and reliable asynchronous processing.
- **Worked in Agile/Scrum teams**, participating in feature development, code reviews, production issue resolution, and continuous delivery through CI/CD pipelines.

Programming Languages, Frameworks and Libraries

1. **Frontend** - React, Vanilla JS, JQuery, React Testing Library, Jest, React Query, Next JS, Astro JS, TypeScript, Transformers.js
2. **Backend** - Nest JS, Express JS, Sequelize, OpenAI, Langchain, Node JS,
3. **HTML & CSS**
4. **UI Libraries:** Tailwind, Antd, ReactStrap, MuI, Bootstrap
5. **State Management :** Redux toolkit, React Query, Redux toolkit Query
6. **AI & LLM:** RAG, Embeddings, Tool-calling, Prompt Engineering, Huggingface, OpenAI, Ollama, chroma, pinecone, vector search, streaming
7. **Databases:** Postgres and Pgvector, Chroma, SQL, MySQL, Pinecone,
8. **Cloud:** AWS EC2, ECS, SQS, SNS
9. **Tools:** Git, Docker, Ollama, Mysql workbench, dbeaver, linux, kubernetes